

Deepanjay Nandal

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SUMMARY

Senior Software Engineer with 3+ years building systems, from CVE validation infrastructure for Google's Android Security team to LLM inference caching and event-driven pipelines processing 100k+ daily events. Combines low-level systems and security depth with AI infrastructure across semantic caching, hybrid retrieval, and Kafka/Redis pipelines.

EXPERIENCE

Discover Excellence LLC

California, USA

Software Engineer (Contract)

Jan 2026 – Jun 2026

- Built a high-throughput Kafka-based visitor intelligence pipeline for an AI-powered eCommerce analytics platform, processing 100k+ daily behavioral events at sub-40ms p95 latency to drive discount outreach via AWS SES.
- Served personalized product feeds at the edge via Cloudflare, backed by Redis collaborative filtering, achieving 92% cache hit rate and sub-30ms p95 latency for multi-tenant deployments.
- Implemented a fault-tolerant BullMQ background pipeline for async job processing, handling 50k+ daily jobs with guaranteed job delivery through persistent state, retries, exponential backoff, and idempotency keys.

Ittiam Systems Pvt Ltd

Bengaluru, India

Senior Software Engineer

Mar 2022 – Jun 2024

- Authored 50+ CTS/STS validation tests merged into the Android Open Source Project (AOSP), each proving a CVE patch for Google's Android Security team, working directly with Google engineers.
- Developed a Ghidra-based binary analysis approach for the Android Bluetooth stack that Google had marked infeasible, which the team adopted to validate 40+ Bluetooth CVEs.
- Engineered a Frida instrumentation framework in Python and Java to hook private Android methods at runtime, enabling the team to reproduce 30+ CVEs that standard testing could not reach.
- Built a GCP Python pipeline ingesting ClusterFuzz crash outputs across Dockerized environments, auto-classifying reports by CVSS severity into a PostgreSQL-backed schema, cutting manual triage overhead by 35%.
- Automated webhook-driven Gerrit CI/CD workflows with Change-Id-keyed idempotency, processing 30–40 CVE submissions per month without duplicate change creation.

PROJECTS

BitMod (AI Inference Cache) | *Python, FastAPI, Vector Embeddings, Multi-tenant, Docker*

Jan – Jun 2026

- Architected a drop-in 9-layer LLM cache proxy with SHA-256 exact lookup, semantic reuse, and Bayesian evidence scoring, achieving 76% hit rate on a 50-query benchmark (100% on repeats, 53% on rephrasings).
- Held zero false serves across 375 held-out queries with nothing cached, reached 90–101x speedup over self-hosted LLM calls at sub-90ms cached, and enforced multi-tenant namespace isolation in SQL.
- Designed cost-aware LRU eviction scoring entries by predicted future hits against original LLM spend.

CaseTally (Legal Search Platform) | *FastAPI, pgvector, RAG, LLM APIs*

Jun – Dec 2025

- Constructed a hybrid legal search system over 83k+ chunks across 47k U.S. Code citations fusing PostgreSQL full-text and pgvector HNSW at p50 18ms and p95 36ms, with LLM query rewriting lifting MRR 10%.
- Instrumented a reproducible retrieval evaluation harness tracking Precision@3, Recall@5, and MRR with relevance derived from citation structure, and streamed answers to the client over FastAPI SSE.
- Engineered an idempotent ingestion pipeline across all 53 U.S. Code titles with an asynchronous embedding worker using FOR UPDATE SKIP LOCKED, treating the NULL embedding column as a crash-safe queue.

WeVibe | *Backend Engineer, Appy.yo (Capstone) | Node.js, PostgreSQL, Socket.IO, Redis*

Jan – Apr 2026

- Created a real-time matchmaking backend using Socket.IO and Redis pub/sub for horizontally scalable event delivery, PostgreSQL advisory locks preventing race conditions, and PostGIS geo-filtering on Google Cloud Run.
- Integrated Gemini bio generation with a custom prompt-injection guard and rate limiting (5 bios/day, 60s cooldown), Firebase JWT auth, and Apple Sign-In with revocation, validated across 17 Jest integration suites.

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, SQL

Backend Technologies: Distributed Systems, Microservices, Event-Driven Architecture, Node.js, Multi-tenancy, Concurrency Control, REST APIs, WebSockets, Message Queues, BullMQ, Rate Limiting, Socket.IO

Data & Storage: Apache Kafka, PostgreSQL, PostGIS, SQLite, Redis, pgvector, Qdrant, HNSW, Distributed Caching

Security & Systems: Android Open Source Project (AOSP), Binary Analysis, Ghidra, Frida, ClusterFuzz

Cloud & DevOps: AWS (EC2, Lambda, SQS, IoT Greengrass, SES), GCP, Firebase, Cloudflare, Docker, CI/CD

AI Infrastructure: Retrieval-Augmented Generation (RAG), Vector Search, Large Language Model (LLM) APIs, Semantic Caching, Embeddings, Collaborative Filtering, Retrieval Evaluation

EDUCATION

Arizona State University

Tempe, Arizona, USA

Master of Science (M.S.) in Computer Software Engineering

Aug 2024 – May 2026

Vellore Institute of Technology

Tamil Nadu, India

Bachelor of Technology (B.Tech) in Information Technology

Jul 2017 – Jul 2021